INTRODUCTION

Molecular biology is a discipline of biology which aims to understand the basic cellular functions such as growth, division, specialization, movement, interaction etc... in terms of the cell's macro molecular components. The term molecular biology was coined in 1938 by Warren Weaver, who was the director of natural sciences department at Rokefeller foundation. Though DNA was identified to be the genetic material as early as 1869, the exact nature of its function was understood only after the discovery of the molecular structure of DNA by J.D Watson and F.H.C Crick in 1952. The concept of base pairing introduced by them directly lead to the understanding DNA replication, RNA synthesis etc. Following these discoveries, the new field of molecular biology progressed rapidly allowing scientists to map out the fundamental biological principles behind macromolecular synthesis and function inside the cell. Today molecular biology has evolved into a very important field of biology with important applications like disease diagnosis and therapy, industrial applications like biotechnology products and to improve agricultural yields.

6.1 THE CENTRAL DOGMA OF MOLECULAR BIOLOGY

The central dogma of molecular biology tries to explain how the genetic information contained in the DNA is conveyed or expressed in the biological system. This dogma was first stated by Crick in 1958 and later revised by Watson in 1965. This describes the detailed residue by residue transfer of the genetic information from DNA to Protein. This dogma is the frame work for understanding the information transfer between the important biological macromolecules: the DNA, RNA and proteins. When cells divide, a new copy of DNA has to be synthesised from the existing DNA. This process where a new DNA molecule is synthesised with the existing DNA as a template is known as replication. The information in the DNA can be copied to RNA by a process called transcription. RNA molecules can act as templates to produce protein molecules by a process called translation.

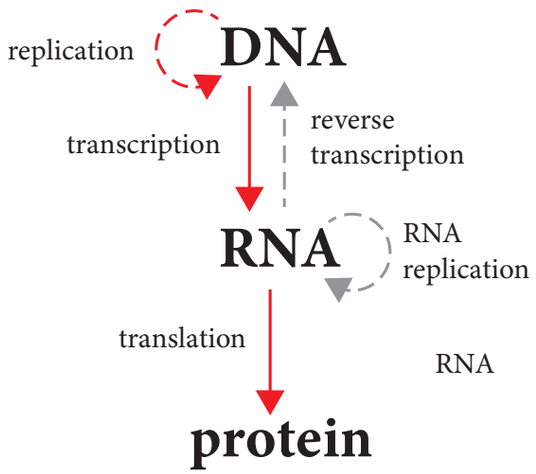

Figure 1: The central dogma of molecular biology

This simple information flow from DNA to RNA to Protein is the common mechanism in most organisms. But rarely, as in the case of retroviruses DNA molecule can be synthesised from RNA molecules (reverse transcription). Similarly RNA molecules are also known to replicate ie synthesise new RNA molecules in RNA viruses such as polio virus and mengoviruses.

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